

Algebraic Fractions

Worksheet 1 (Non-Calculator)

1. **S15/qp11/q11**

Express $\frac{1}{x} + 2 - \frac{3}{x+1}$ as a single fraction in its simplest form. [3]

2. **W16/qp12/q20**

Solve the equation $\frac{2x-1}{4} + \frac{x-2}{3} = 2$. [3]

3. **S17/qp12/q23(a)**

Solve $\frac{7x}{4-3x} = 3$ [2]

4. **S17/qp21/q22 (0580)**

Write as a single fraction in its simplest form.

a. $\frac{x^2-3x}{x^2-9}$ [3]

b. $\frac{3}{x-4} + \frac{2}{2x+5}$ [3]

5. **S17/qp22/q20 (0580)**

Write as a single fraction in its simplest form.

$$\frac{2x-1}{3} + \frac{2}{x+1}$$

[3]

6. **W17/qp21/q19 (0580)**

Write as a single fraction in its simplest form.

$$\frac{5}{x-3} + \frac{3}{x+7} + \frac{1}{2}$$

[4]

7. S18/qp12/q8

Solve

$$\frac{4}{x-11} = \frac{1}{3x}$$

[2]

8. S18/qp12/q9

Express each of the following as a single fraction in its simplest form.

a. $\frac{2}{3a} + \frac{5}{2a}$ [1]

b. $\frac{5}{2b^2} \div \frac{15}{4b^3}$ [2]

9. S20/qp11/q26

Write as a single fraction in its simplest form.

$$\frac{2x+3}{x+4} - \frac{5}{3x-2}$$

[4]

10. S20/qp12/q23

Express as a single fraction in its simplest form.

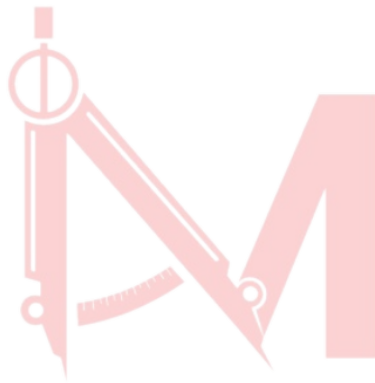
$$\frac{5}{2x-1} - \frac{3}{x+4}$$

[3]

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Answer Key:

1. $\frac{2x^2+1}{x(x+1)}$
2. 3.5
3. $\frac{3}{4}$
4. (a) $\frac{x}{x+3}$ (b) $\frac{8x+7}{(x-4)(2x+5)}$
5. $\frac{2x^2+x-7}{3x+3}$
6. $\frac{x^2+20x+31}{2(x-3)(x+7)}$
7. -1
8. (a) $\frac{19}{6a}$ (b) $\frac{2b}{3}$
9. $\frac{6x^2-26}{(x+4)(3x-2)}$
10. $\frac{23-x}{(2x-1)(x+4)}$



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